NF TAPE MACHINE

ANALOG TAPE EMULATOR — USER MANUAL

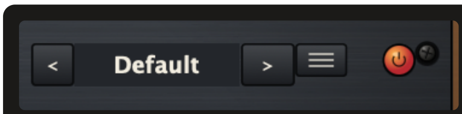
Version 0.1 · NF Audio Tools · VST3 / AU / Standalone

Contents

1. Overview

NF Tape Machine is a clean-room analog tape emulator built around four classic tape-formula profiles, three transport speeds and two reproduce-head curves. It reproduces the character of a real reel-to-reel deck — tape saturation, head-bump low end, high-frequency roll-off, wow & flutter, hiss and dropouts — all under direct control rather than baked into a single "vibe" knob.

The plugin is designed to be usable at two extremes: completely transparent (drive low, character sources switched out) for mix-bus glue, and deliberately worn and characterful for creative sound design — with the Default preset sitting deliberately close to the transparent end, so inserting the plugin doesn't surprise you.



PRESET BAR & POWER

2. Installation

macOS

1. Open `NF Tape Machine – Mac Installer.dmg`.
2. Double-click `NF Tape Machine Installer.pkg` and follow the prompts.
3. The installer places the plugin in:
   - `/Library/Audio/Plug-Ins/VST3/NF Tape Machine.vst3`
   - `/Library/Audio/Plug-Ins/Components/NF Tape Machine.component` (Audio Unit)
4. Rescan plugins in your DAW if it doesn't appear automatically.

Windows

1. Run `NF Tape Machine Installer.exe`.
2. The installer places the VST3 in `C:\Program Files\Common Files\VST3\NF Tape Machine.vst3`.
3. Rescan plugins in your DAW if it doesn't appear automatically.

> An uninstaller is included on both platforms (macOS: re-run the package and choose remove where offered; Windows: "Uninstall NF Tape Machine" from Control Panel / Apps).

3. Signal Flow

Input Trim → High-Pass Filter → Tape Saturation (Drive / Bias) → Head Bump & HF Roll-off (set by Tape Type, Speed & Repro Head) → Wow & Flutter → Noise / Dropouts → Tone (EQ Low/High) → Low-Pass Filter → Output Trim → Dry/Wet Mix.

Gain Link, when engaged, couples the Input and Output trims so raising one attenuates the other by the same amount in dB — useful for driving the saturation stage harder without changing the overall level hitting your meters.

4. Controls Reference

Input Section

INPUT + HPF

Control	Range	Description
INPUT	-24 to +24 dB	Trim applied before the tape stage. Higher settings drive the saturation harder.
HPF	20 Hz – 200 Hz	High-pass filter ahead of the tape stage, useful for tightening low end before it hits saturation.

Tape Type

TAPE TYPE

Formula	Character
GP9	Modern, high-headroom formula — the most neutral, extended top end.
456	Classic studio formula — warm mids, gentle compression.
499	Punchy, slightly brighter formula with a firmer head bump.
250	Vintage/low-output formula — softer highs, more visible wow & flutter and noise floor.

Drive / Saturation

DRIVE/SAT + BIAS/CAL

Control	Range	Description
DRIVE	0 – 10	Amount of tape saturation. Also drives the tube lamps' glow in the interface.
SAT (IN)	On/Off	Enables/disables the saturation stage entirely.
BIAS	0 – 10	Simulated tape bias — shifts saturation symmetry and harmonic balance.
CAL	On/Off	Bias Calibrated — engages the factory-calibrated bias curve for the selected Tape Type.

Wow & Flutter

WOW & FLUTTER

Control	Range	Description
RATE	0.1 – 10 Hz	Modulation speed of the pitch instability.
DEPTH	0 – 100	Modulation depth. Kept low by default for a subtle "alive" quality rather than audible detuning.
IN	On/Off	Enables/disables wow & flutter. On by default, subtle.

Noise & Dropouts

Control	Range	Description
NOISE	0 – 10	Tape hiss level.
IN (Noise)	On/Off	Enables/disables hiss. Off by default.
DROPOUTS	0 – 10	Frequency/severity of momentary level dropouts.
IN (Dropouts)	On/Off	Enables/disables dropouts. Off by default.

EQ & Gain Link

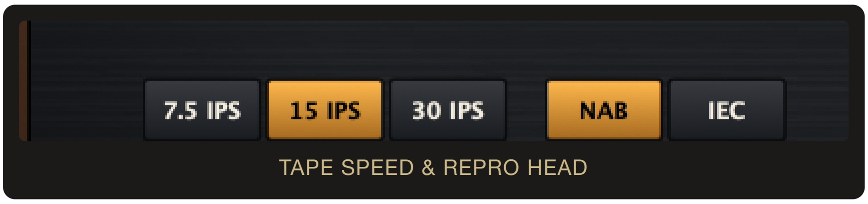
Control	Range	Description
LF	-12 to +12 dB	Low-shelf tone control.
HF	-12 to +12 dB	High-shelf tone control.

Output Section

Control	Range	Description
OUTPUT	-24 to +24 dB	Trim applied after the tape stage.
LPF	2 kHz – 20 kHz	Low-pass filter after the tape stage, for extra high-frequency taming.
LINK	On/Off	Gain Link — see below.

Gain Link couples INPUT and OUTPUT so the two move inversely, dB for dB, keeping overall level roughly constant while you push the saturation harder or gentler. While linked, dragging either knob shows its usual value bubble plus a matching one over the other knob, so both linked values stay visible at once. A single click (no drag) on either the INPUT or OUTPUT knob resets both to 0 dB. Turning Link off frees both knobs to move independently again.

Tape Speed & Repro Head



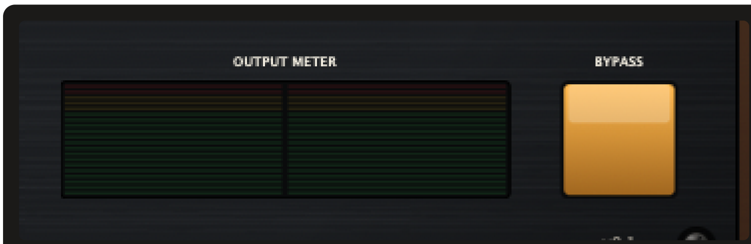
Control	Options	Description
TAPE SPEED	7.5 / 15 / 30 IPS	Higher speeds extend high-frequency response and reduce wow & flutter/noise; lower speeds are darker and more colored.
REPRO HEAD	NAB / IEC	Playback equalization standard — changes the tonal balance of the head-bump/roll-off curve.

Tape Age & Mix

TAPE AGE, DROPOUTS & MIX

Control	Range	Description
TAPE AGE	0 (New) – 100 (Worn)	Global "wear" control — gradually softens highs and increases noticeable imperfections as it's raised.
MIX	0 (Dry) – 100 (Wet)	Blends the processed signal with the untouched input. 100% by default for traditional insert use.

Global



OUTPUT METER & BYPASS

Control	Description
BYPASS	Bypasses the entire tape engine, passing audio through unprocessed.

5. Reels, Tube Lamps & Metering

REELS, VU METERS & NAMEPLATE

Reel-to-reel transport

The two reels are cosmetic but interactive — clicking either one starts or stops the spinning transport for both (they're threaded on the same tape). Each Tape Type recolors the reel finish (gold for GP9, silver for 456, red for 499, black for 250) so you always know which formula is loaded at a glance. They also spin faster or slower to match the selected TAPE SPEED — quickest at 30 IPS, slowest at 7.5 IPS.

Tube lamps

The two lamps flanking the reels brighten and intensify in color as DRIVE is raised, and hold a fixed, vivid resting glow when SAT is switched off or the plugin is bypassed — a visual read on how hard the tape stage is working.

VU meters & output meter

The twin VU dials show the processed signal's level on a standard -20…+3 dB analog scale. The LED-style bar meter below the control rows shows the same left/right output levels in a more modern peak-reading format.

6. Factory Presets

Preset	Character
Default	Subtle, transparent-leaning starting point — saturation and wow & flutter present but restrained, noise and dropouts off.
Warm Bus Glue	Gentle 456-formula warmth and compression for mix-bus duty.
Vintage Slap	499 formula pushed harder, with noise and dropouts engaged for an overtly vintage character.
Mastering Sheen	Very light touch on 499 formula with IEC repro head — for mastering-chain use.
Lo-Fi Worn	250 formula driven hard, high tape age, dropouts and noise all engaged — heavily degraded, lo-fi character.

Use the **<** / **>** arrows either side of the preset name to step through the factory bank.

7. Usage Tips

- For mix-bus glue, keep DRIVE low (2–4), MIX around 100%, and leave NOISE/DROPOUTS off.
- Engage GAIN LINK before dialing in DRIVE so you can judge the saturation's tone without a level jump confusing the comparison — click either knob to snap back to 0/0 and start again.
- Raising TAPE AGE is a fast way to dial in "worn" character without touching every individual parameter by hand.
- Switch TAPE SPEED to 7.5 IPS for a noticeably darker, more colored result; 30 IPS for the cleanest, most extended top end.
- WOW & FLUTTER's DEPTH is intentionally subtle by default — raise it further only when you want an obviously wobbly, lo-fi pitch effect.

8. Credits & Version

NF Tape Machine v0.1 — NF Audio Tools.
Built with JUCE. VST3, AU and Standalone formats.